

Nigeria’s Solid Minerals Sector

Export Dynamics and Global Positioning: A Snapshot Analysis (2022–2023)

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Data sources	NEITI Annual Reports 2022 & 2023; BGS World Mineral Production 2019–2023
Minerals covered	18 selected minerals including lithium, lead/zinc ore, columbite, limestone, tin ore, and others
Period	2022–2023 (two-year snapshot)
Tools used	Microsoft Excel
Key KPIs	Export Intensity (EI) · Global Share · Year-on-Year production and export change

The Headline

Nigeria’s solid minerals sector is extracting and exporting minerals that the world increasingly wants — lithium demand is up over 30% globally, and Nigeria’s lithium production rose over 50% between 2022 and 2023.

But Nigeria participates in these markets without influencing them. High export intensity in lithium, lead/zinc ore, and columbite masks a fundamental structural weakness: Nigeria is a price-taker, not a price-maker. Its global market share remains negligible or low across most minerals, and almost no downstream processing happens before export.

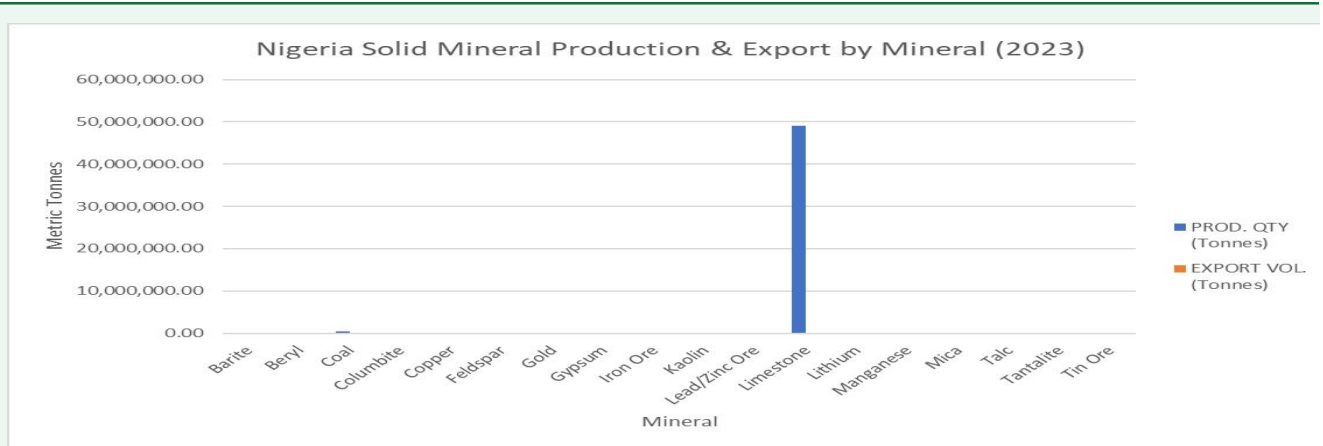
The opportunity is not more extraction. It is more processing.

Production Structure: A Tale of Two Sectors

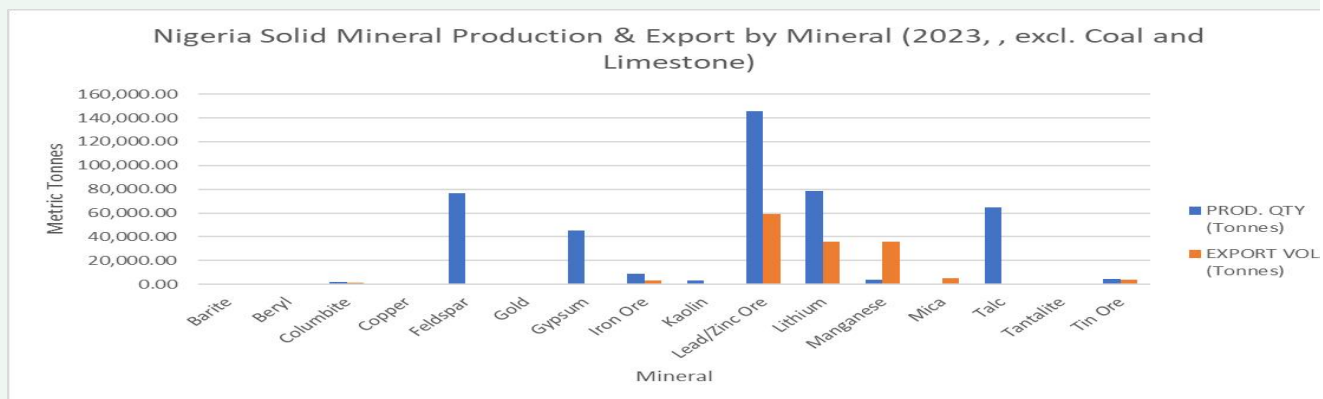
Nigeria’s mineral production is dominated by a single commodity that tells almost nothing about the sector’s export dynamics. Limestone production in 2023 exceeded 49 million metric tonnes, dwarfing every other mineral in the dataset combined. Yet limestone generates almost no export

volume. Its production exists entirely to serve Nigeria's domestic infrastructure and construction needs, reflected in a near-zero export volume.

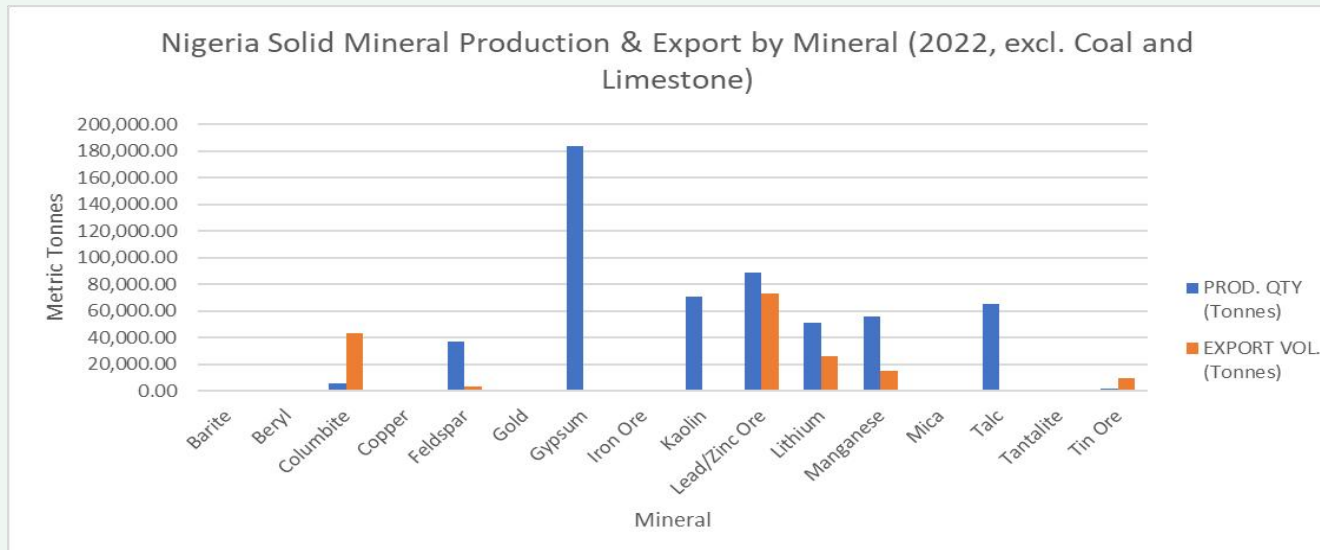
This creates a structural illusion: aggregate production figures look large, but strip out limestone (and coal, the second-largest by volume) and the export-relevant sector is a fraction of that size. The 16 remaining minerals collectively represent a much smaller production base, but they are the ones with genuine global market relevance.



Graph 1: Nigeria Solid Mineral Production and Export by Mineral (2023)



Graph 2: Nigeria Solid Mineral Production and Export by Mineral (2023, Excluding Coal and Limestone)



Graph 3: Nigeria Solid Mineral Production and Export by Mineral (2022, Excluding Coal and Limestone)

Note: Limestone's scale dominates the chart. Graphs 2 and 3 (excluding coal and limestone) reveal the structure of the export-relevant sector.

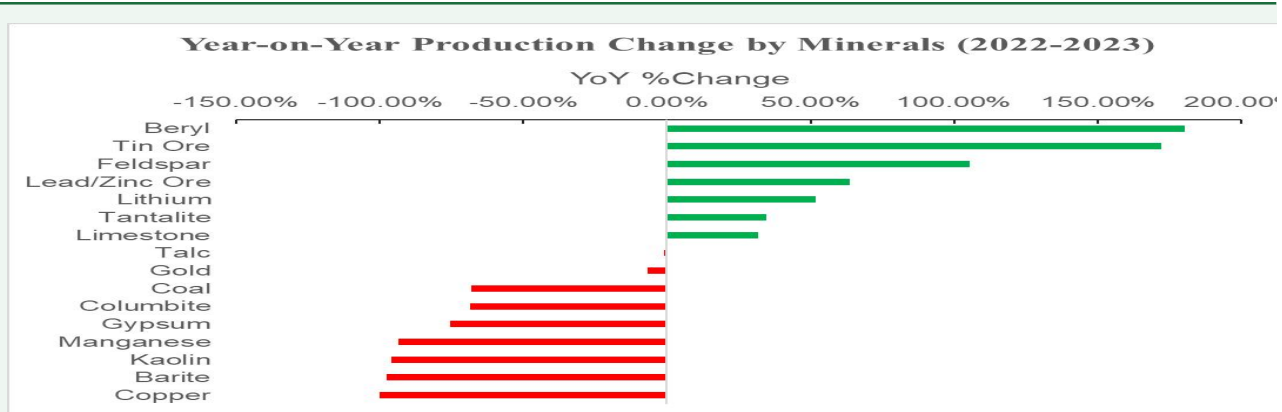
Across the export-relevant minerals, a clear split emerges between domestically absorbed commodities and export-oriented ones. Gypsum, feldspar, kaolin, and talc are predominantly consumed within Nigeria. Lead/zinc ore, lithium, manganese, and columbite, in contrast, send a large share, in some cases more than they produce in a given year, to international markets.

That last point is important. Several minerals show Export Intensity values above 1.0, meaning recorded exports exceeded recorded production for that year. This does not indicate a data error. It reflects stockpile drawdowns, reporting timing differences between production and export records, or re-export activity. These figures should be read as indicators of strong export orientation, not as precise real-time production-to-export ratios.

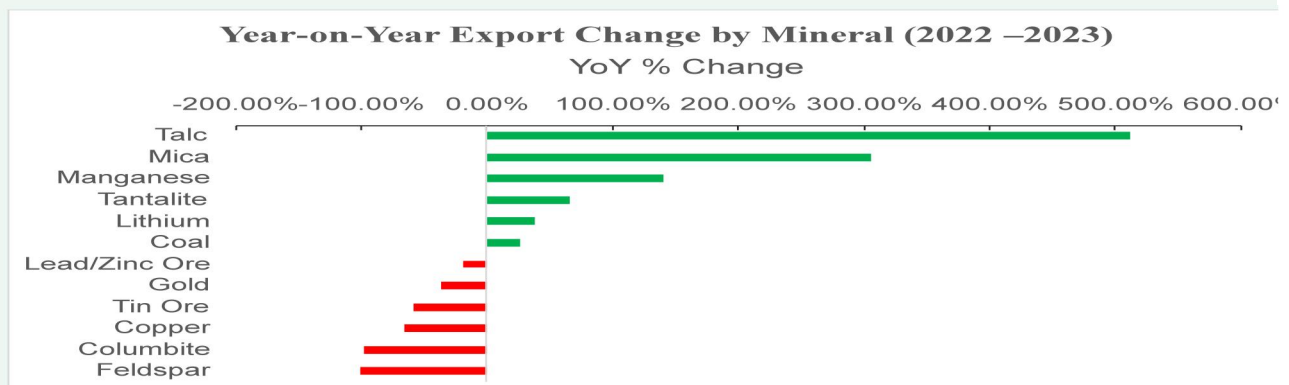
Responsiveness to Global Demand

One of the clearest findings from this analysis is that Nigeria's minerals sector is not operating in isolation. It is responding, rapidly in some cases, to shifts in global demand.

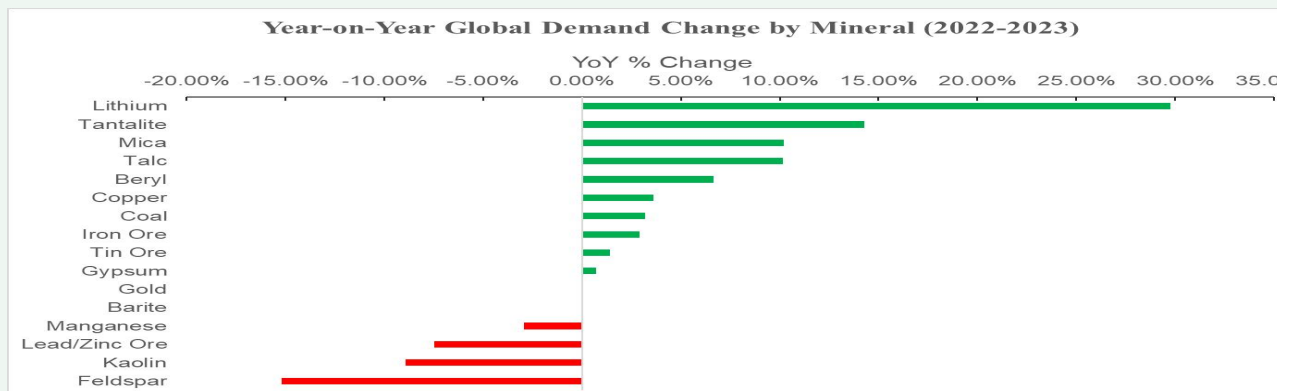
Among the five minerals with the fastest-growing global demand between 2022 and 2023 - lithium, tantalite, mica, talc, and beryl - Nigeria's production responded positively in three cases. Lithium is the most striking: global demand rose approximately 30%, and Nigeria's lithium production increased by over 50% in the same period. Export volumes followed, rising significantly year-on-year.



Graph 4: Year-on-Year Production Change by Minerals (2022-2023)



Graph 5: Year-on-Year Export Change by Minerals (2022-2023)



Graph 6: Year-on-Year Global Demand Change by Minerals (2022-2023)

Beryl is a notable case within this group. Nigeria exported zero beryl in 2022, then shipped 189 tonnes in 2023 — an emergence from zero, rather than growth from a prior base. This pattern suggests that new or informal mining activity is being formalised or activated in response to global price signals.

These patterns collectively suggest an export-oriented segment of the Nigerian minerals sector that is actively tracking and responding to global commodity demand. That is a meaningful commercial signal. But it also raises a follow-on question the data cannot yet answer: are Nigerian miners and exporters capturing the price upside of these demand surges, or simply supplying into them at whatever price global buyers offer?

Given Nigeria’s low global market share across most of these minerals (see Section 4), the answer is almost certainly the latter.

Responsiveness to demand is not the same as market power. Nigeria is filling orders, not setting prices.

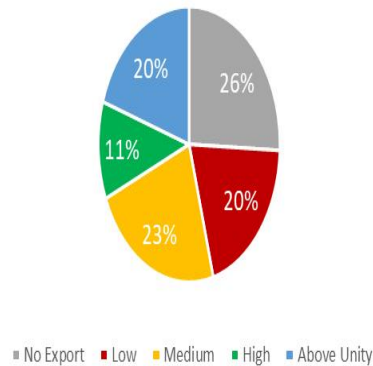
Export Intensity: Who Is Mining for Nigeria, and Who for the World?

Export Intensity (EI) provides the clearest picture of which minerals are domestically driven and which are globally oriented. The derived KPI, the ratio of export volume to production volume for each mineral and year, segments Nigeria’s minerals into five categories.

KPI	Definition	Scale / Categories
Export Intensity (EI)	Export volume ÷ Production volume for a given mineral and year	No Export (0) · Low (<0.25) · Medium (0.25–0.75) · High (0.75–1) · Above Unity (>1)
Global Share	Nigeria’s production ÷ Global production for that mineral	Negligible (<0.001%) · Low (0.01–0.1%) · Moderate (0.1–1%) · Significant (>1%)

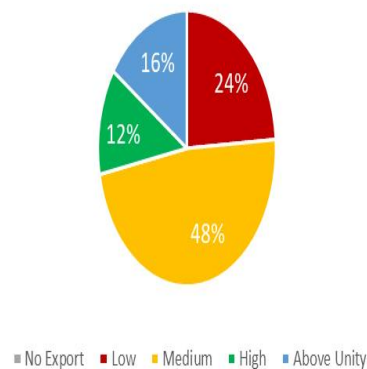
Counted by number of minerals, the largest single category across 2022 and 2023 is ‘No Export’ at 26%, minerals being mined entirely for domestic consumption. But the distribution across the other categories is surprisingly even, with High and Above-Unity intensity minerals representing a combined 31% by count.

Distribution of Minerals by Intensity Category
(2022-2023)



Graph 7

Distribution of Total Export Volume by Intensity Category (2022-2023)



Graph 8

When measured by export volume rather than mineral count, the picture shifts. Medium-intensity minerals dominate the total volume exported — accounting for 48% of all export tonnes. This is largely explained by lead/zinc ore, which combines high absolute volumes with a medium EI ratio. Above-unity minerals, mica, manganese, and columbite in certain years contribute a further 24% of export volume despite representing only a few minerals.

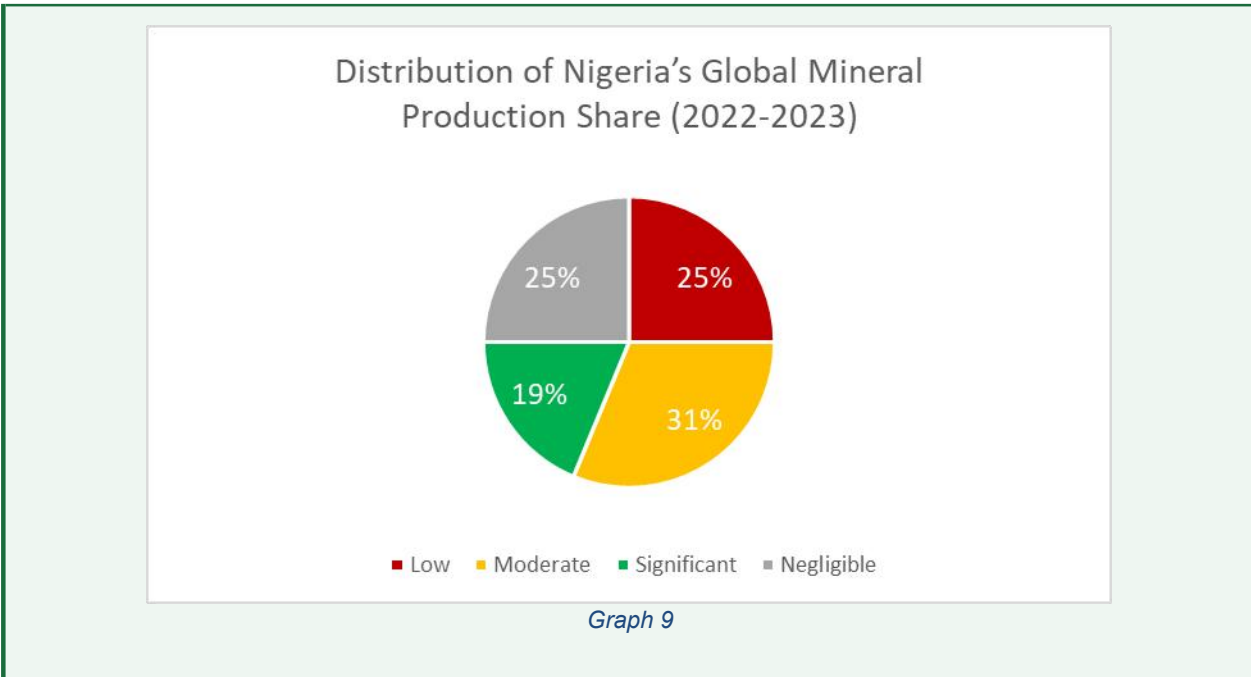
The practical implication: Nigeria's export base is neither uniformly domestic nor uniformly export-oriented. It contains a subset of minerals that are aggressively export-focused, sitting

alongside a larger group that serves the Nigerian market. Understanding which minerals sit in which category, and why, is essential for targeted investment and policy decisions.

Global Market Share: Participating Without Influencing

This is where the analysis becomes most important for strategic planning. Export intensity tells us how much of Nigeria’s production leaves the country. Global share tells us how much Nigeria actually matters to the world’s supply of each mineral. The two metrics together reveal a critical gap.

Only 19% of Nigeria’s mineral production slots across 2022 and 2023 fall into the ‘Significant’ global share category — defined as a production share above 1% of global output. The minerals in this category are lithium, beryl, and tin ore. Nigeria’s lithium production share in 2023 was approximately 37% of global output, a genuinely significant presence. Tin ore sits at around 16%. These are real positions.



But the majority of minerals, including those with high export intensity like manganese and columbite, have low or negligible global market share. Manganese exports surged in 2023, yet Nigeria’s global production share remains under 0.01%. Columbite, for which Nigeria is historically significant, shows no global share figure in this dataset due to data alignment gaps, but prior evidence suggests it too remains a modest global contributor.

The structural gap: several minerals are export-oriented without being globally influential.

Nigeria sends a large share of its manganese, mica, and columbite production abroad, but the absolute volumes are small enough that Nigeria's presence does not move global prices or constrain global supply. It participates in these markets as a peripheral supplier, not as a market-shaping producer.

The Central Problem: Extraction Without Value Capture

Every thread in this analysis leads to the same conclusion. Nigeria mines minerals the world wants. It exports them at scale. It responds to global demand signals. But it does not process them before export, does not capture downstream value, and does not hold enough market share in most commodities to influence the prices it receives.

This is not unique to Nigeria; it is the defining challenge of resource-dependent economies across Sub-Saharan Africa. But it is particularly acute in a moment when the minerals Nigeria holds, lithium, tantalite, tin, and others tied to the global energy transition, are rising sharply in value at the processed and refined stage, not the raw ore stage.

Consider lithium. Nigeria's raw lithium ore exports in 2023 generated revenue at spodumene concentrate prices, roughly \$700–\$1,500 per tonne at the time of this analysis. Battery-grade lithium carbonate (one processing step further) was trading at multiples of that figure. The value is not in the rock. It is in what is done to the rock before it leaves the country.

Columbite presents a similar case. Nigeria was historically one of the world's leading columbite producers. The mineral is a source of niobium, a critical alloying element in high-strength steel and superconducting materials. Refined niobium commands significant premiums over raw columbite. Yet Nigeria exports ore, not metal.

Conclusion

Nigeria's solid minerals sector in 2022 and 2023 shows genuine commercial activity, meaningful responsiveness to global demand, and a clear export orientation in several strategic minerals. These are the foundations of a sector with real potential.

But the analysis also reveals that potential is not being captured. Nigeria's role in global mineral markets is primarily as a raw material supplier — participating in commodity price cycles it does not influence, exporting value it does not retain, and operating with data infrastructure that makes even basic performance measurement difficult.

Nigeria is selling the ingredients while the world profits from the recipe.

Policy and Investment Implications

1. Prioritise downstream processing for lithium, tin, and columbite where Nigeria already has meaningful production scale and rising global demand provides a commercial window.
2. Improve data infrastructure. The two-year window of this analysis, combined with classification inconsistencies between NEITI and BGS datasets, limits strategic decision-making. A national minerals data system with standardised, annual reporting is a prerequisite for evidence-based sector management.
3. Diversify export-oriented minerals beyond the current concentration in lead/zinc ore and lithium. Tantalite and beryl show emerging export activity and rising global demand — both merit structured development.
4. Strengthen integration into global value chains through infrastructure investment, regulatory clarity, and targeted incentives for beneficiation. Market share gains require not just more mining, but credible supply-chain positioning.

Recommended Further Analyses

- Price and value analysis: unit value per tonne exported by mineral, tracked against global commodity prices
- Firm-level and mine-level data: production efficiency, cost structures, and operator profiles
- Beneficiation potential assessment: which minerals have the highest value-add margin between ore and processed output in the Nigerian context?
- Longer time-series analysis: 2018–2023 to assess structural trends rather than snapshot patterns

- Comparative analysis: Nigeria’s export intensity and global share versus peer African producers (DRC, Ghana, Zimbabwe) in shared minerals

Limitations and Data Notes

This analysis was conducted with care to accurately reflect available data. The following limitations should be noted when interpreting findings:

- Two-year data window (2022–2023 only): restricts long-term trend analysis and limits the robustness of year-on-year comparisons
- Classification mismatches between NEITI and BGS datasets: lead and zinc are reported jointly in NEITI but separately in WMP; aggregation methodology introduces potential inaccuracy in global comparison figures
- No price or value-based metrics: all analysis is volume-based; economic value, profitability, and margin dynamics are not captured
- No firm-level or mine-level data: findings reflect aggregate sector performance, not individual operator dynamics
- Above-unity Export Intensity values: interpreted as export orientation indicators, not precise real-time ratios; likely reflect stockpile drawdowns, reporting timing gaps, or re-export activity